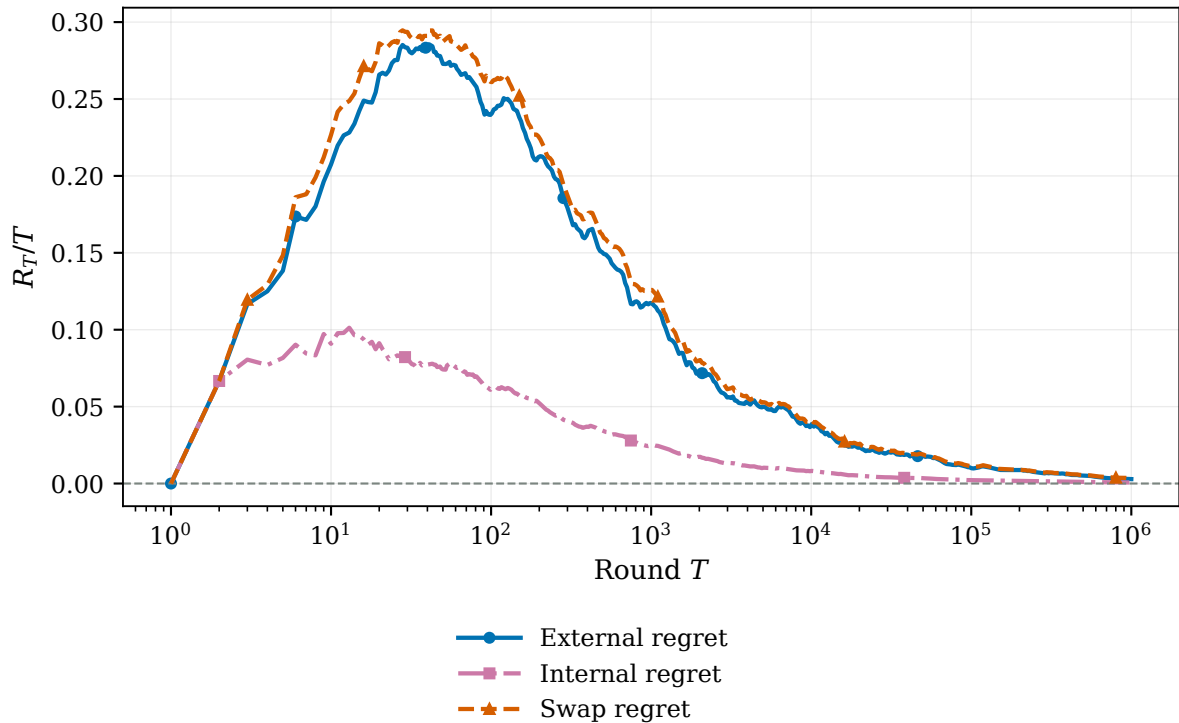


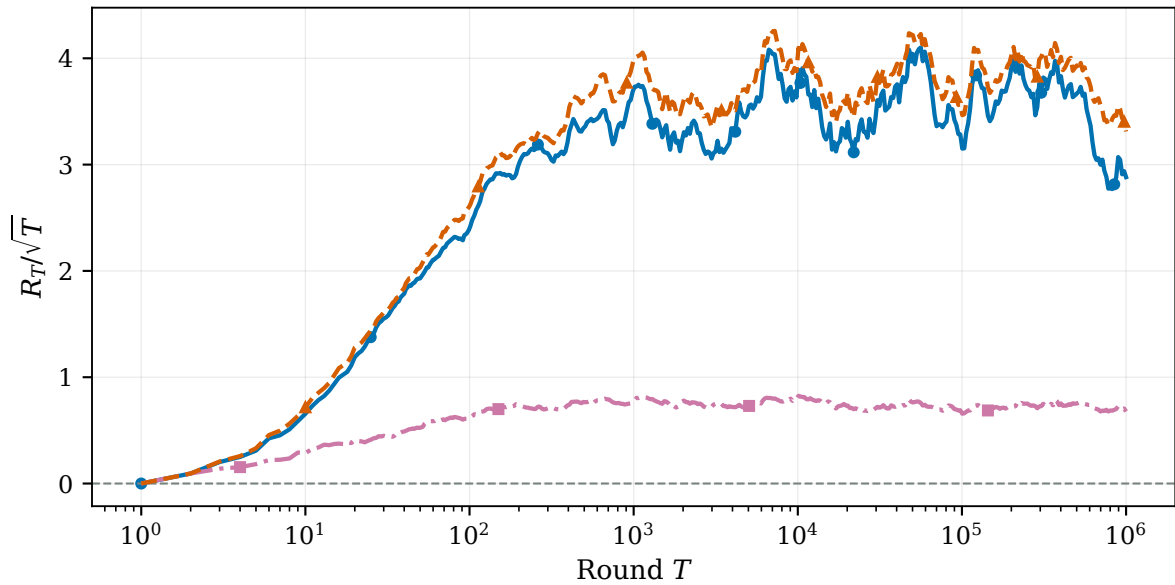
Experiment Information

Figure: Regret-notion comparison
Environment: Lazy random walk
Actions: 10
Feedback: Bandit feedback
Algorithm: EXP3
Horizon: 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External: mean = 0.0028822, SD = 0.000932987, SE = 0.00026933
Internal: mean = 0.000677917, SD = 0.000204923, SE = 5.91562e-05
Swap: mean = 0.0033119, SD = 0.000741822, SE = 0.000214146



Experiment Information

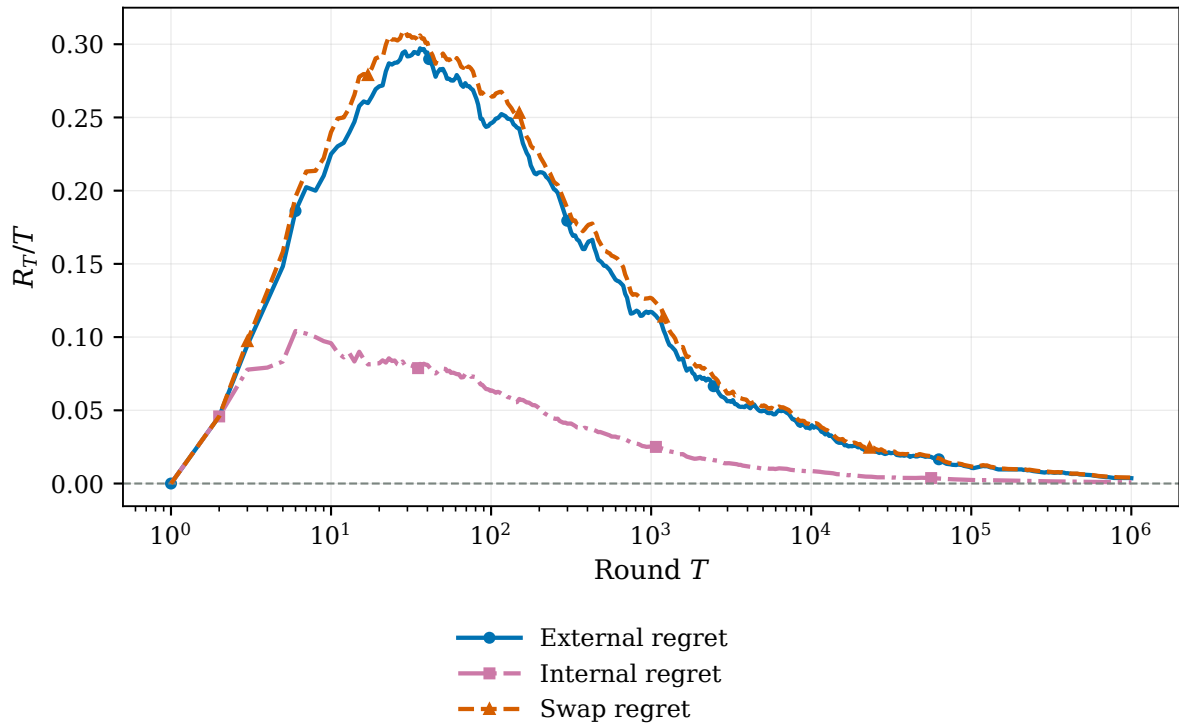
Figure:	Regret-notion comparison
Environment:	Lazy random walk
Actions:	10
Feedback:	Bandit feedback
Algorithm:	EXP3
Horizon:	1,000,000
Replicates:	12
Seed:	42
Regrets:	External, Internal, Swap
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External:	mean = 2.8822, SD = 0.932987, SE = 0.26933
Internal:	mean = 0.677917, SD = 0.204923, SE = 0.0591562
Swap:	mean = 3.3119, SD = 0.741822, SE = 0.214146



- External regret
- Internal regret
- Swap regret

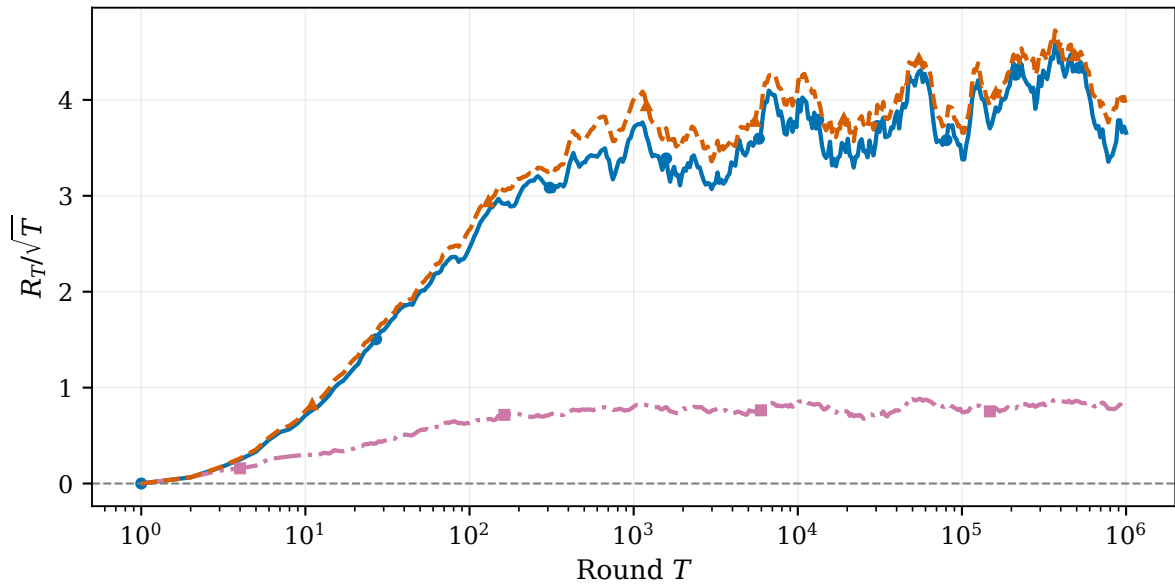
Experiment Information

Figure:	Regret-notion comparison
Environment:	Lazy random walk
Actions:	10
Feedback:	Bandit feedback
Algorithm:	BM-EXP3
Horizon:	1,000,000
Replicates:	12
Seed:	42
Regrets:	External, Internal, Swap
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External:	mean = 0.00365742, SD = 0.00150648, SE = 0.000434883
Internal:	mean = 0.000806575, SD = 0.000220401, SE = 6.36244e-05
Swap:	mean = 0.00394529, SD = 0.00136933, SE = 0.000395292



Experiment Information

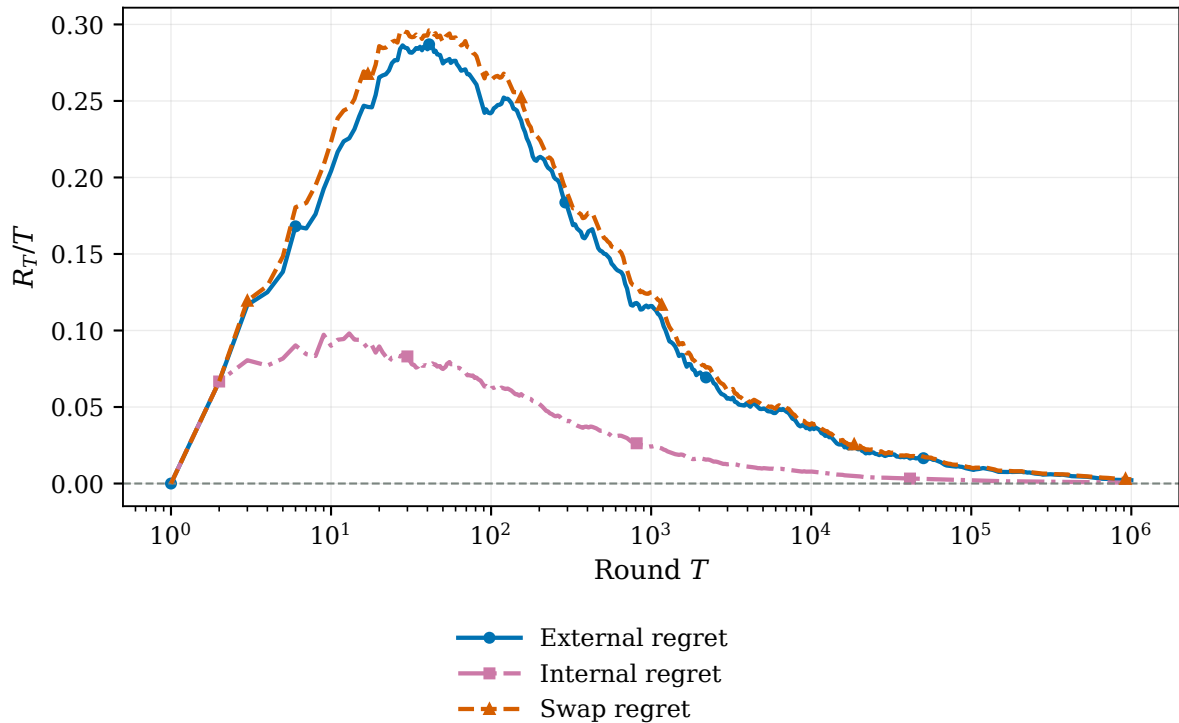
Figure:	Regret-notion comparison
Environment:	Lazy random walk
Actions:	10
Feedback:	Bandit feedback
Algorithm:	BM-EXP3
Horizon:	1,000,000
Replicates:	12
Seed:	42
Regrets:	External, Internal, Swap
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External:	mean = 3.65742, SD = 1.50648, SE = 0.434883
Internal:	mean = 0.806575, SD = 0.220401, SE = 0.0636244
Swap:	mean = 3.94529, SD = 1.36933, SE = 0.395292



- External regret
- Internal regret
- Swap regret

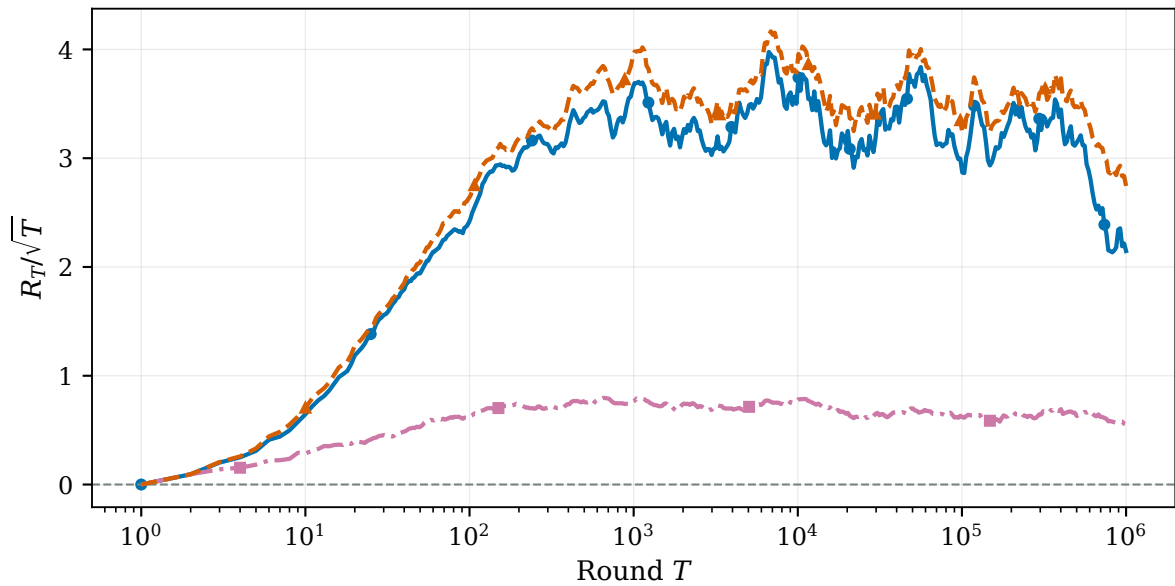
Experiment Information

Figure:	Regret-notion comparison
Environment:	Lazy random walk
Actions:	10
Feedback:	Bandit feedback
Algorithm:	EXP3-IX
Horizon:	1,000,000
Replicates:	12
Seed:	42
Regrets:	External, Internal, Swap
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External:	mean = 0.002147, SD = 0.00113197, SE = 0.00032677
Internal:	mean = 0.000552792, SD = 0.00013908, SE = 4.01491e-05
Swap:	mean = 0.00274492, SD = 0.000825438, SE = 0.000238283



Experiment Information

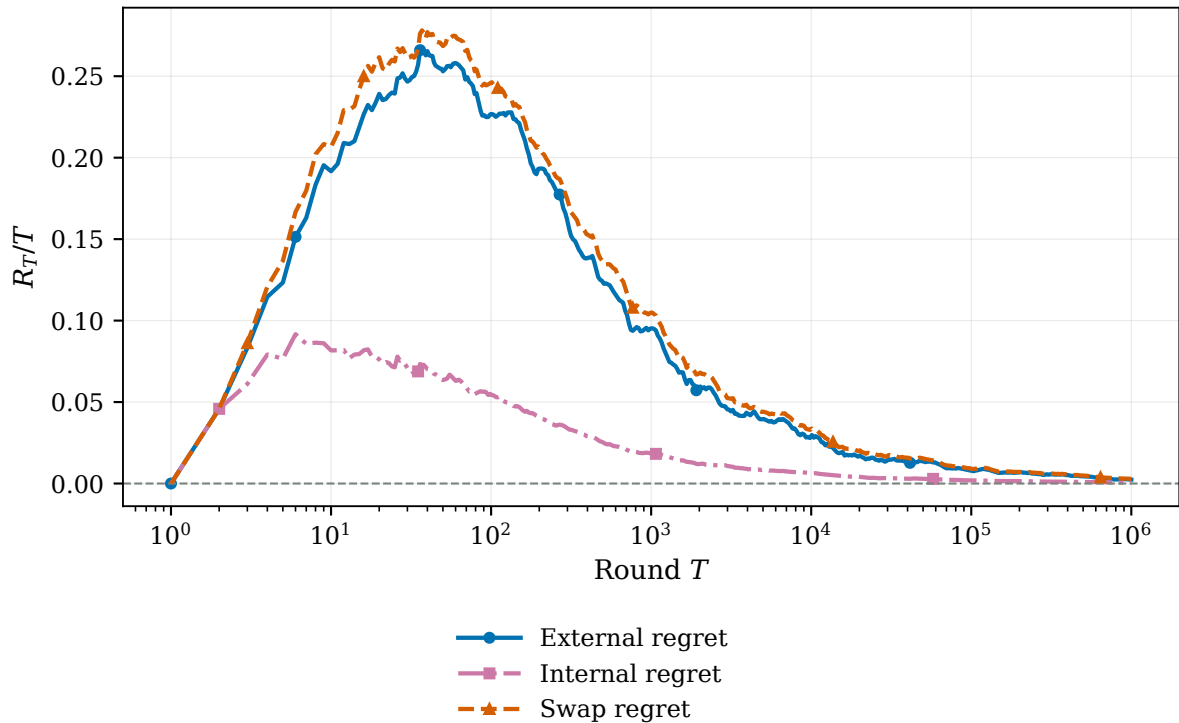
Figure:	Regret-notion comparison
Environment:	Lazy random walk
Actions:	10
Feedback:	Bandit feedback
Algorithm:	EXP3-IX
Horizon:	1,000,000
Replicates:	12
Seed:	42
Regrets:	External, Internal, Swap
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External:	mean = 2.147, SD = 1.13197, SE = 0.32677
Internal:	mean = 0.552792, SD = 0.13908, SE = 0.0401491
Swap:	mean = 2.74492, SD = 0.825438, SE = 0.238283



- External regret
- Internal regret
- Swap regret

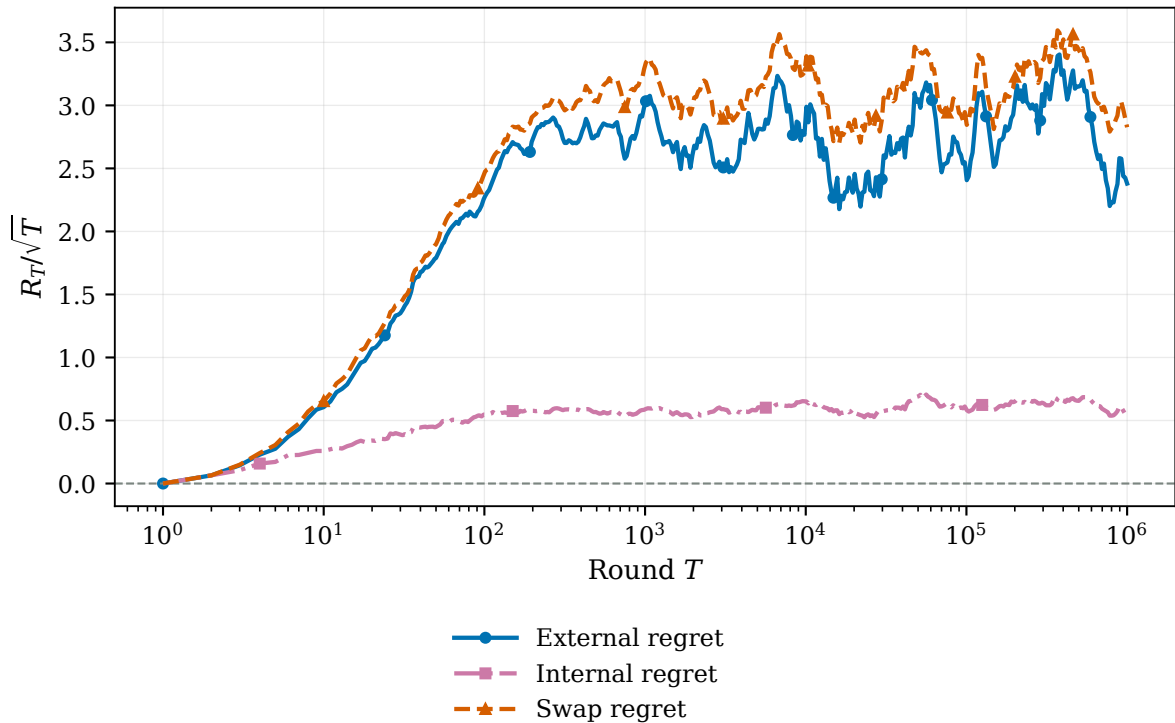
Experiment Information

Figure:	Regret-notion comparison
Environment:	Lazy random walk
Actions:	10
Feedback:	Bandit feedback
Algorithm:	Ito-Tsallis
Horizon:	1,000,000
Replicates:	12
Seed:	42
Regrets:	External, Internal, Swap
View:	R / T
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External:	mean = 0.00237984, SD = 0.00127608, SE = 0.000368372
Internal:	mean = 0.00054095, SD = 0.000133307, SE = 3.84825e-05
Swap:	mean = 0.00282644, SD = 0.0010513, SE = 0.000303485



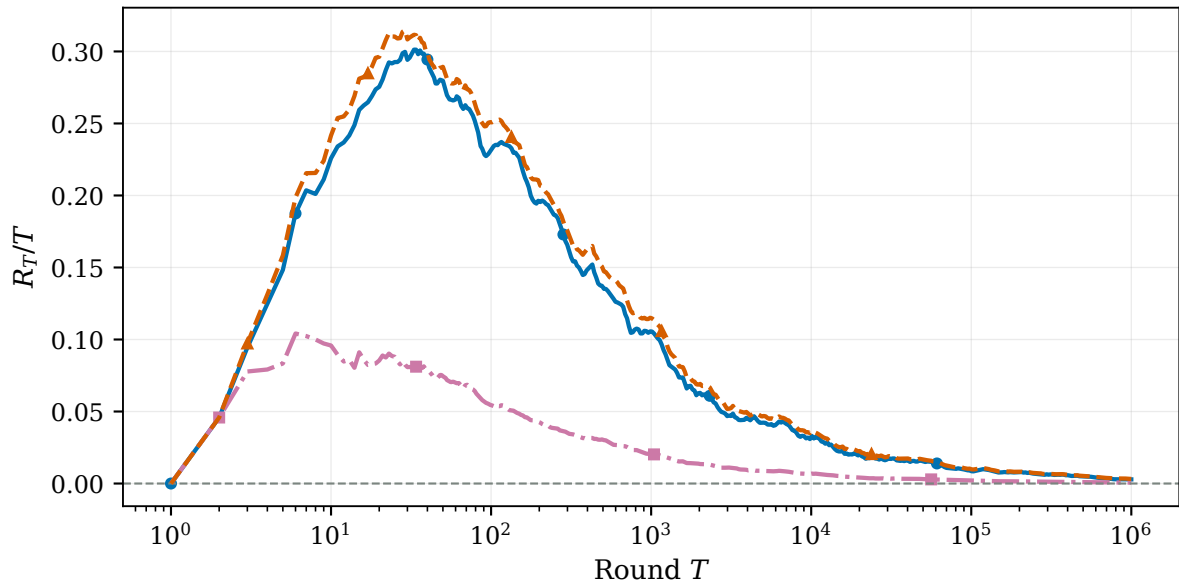
Experiment Information

Figure:	Regret-notion comparison
Environment:	Lazy random walk
Actions:	10
Feedback:	Bandit feedback
Algorithm:	Ito-Tsallis
Horizon:	1,000,000
Replicates:	12
Seed:	42
Regrets:	External, Internal, Swap
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External:	mean = 2.37984, SD = 1.27608, SE = 0.368372
Internal:	mean = 0.54095, SD = 0.133307, SE = 0.0384825
Swap:	mean = 2.82644, SD = 1.0513, SE = 0.303485



Experiment Information

Figure: Regret-notion comparison
Environment: Lazy random walk
Actions: 10
Feedback: Bandit feedback
Algorithm: LCE-IX
Horizon: 1,000,000
Replicates: 12
Seed: 42
Regrets: External, Internal, Swap
View: R / T
Final endpoints at T: 1,000,000
Endpoint statistics: replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External: mean = 0.00287459, SD = 0.00137781, SE = 0.000397738
Internal: mean = 0.000649375, SD = 0.000157445, SE = 4.54504e-05
Swap: mean = 0.00317503, SD = 0.00120085, SE = 0.000346655



- External regret
- Internal regret
- Swap regret

Experiment Information

Figure:	Regret-notion comparison
Environment:	Lazy random walk
Actions:	10
Feedback:	Bandit feedback
Algorithm:	LCE-IX
Horizon:	1,000,000
Replicates:	12
Seed:	42
Regrets:	External, Internal, Swap
View:	$R / \sqrt{T}$
Final endpoints at T:	1,000,000
Endpoint statistics:	replicate mean, sample SD (ddof=1), SE=SD/sqrt(n)
External:	mean = 2.87459, SD = 1.37781, SE = 0.397738
Internal:	mean = 0.649375, SD = 0.157445, SE = 0.0454504
Swap:	mean = 3.17503, SD = 1.20085, SE = 0.346655

